

Conditional probability, independence and Bayes' rule

Condition on what you know: keep only the outcomes in B and rescale. Bayes' rule turns $P(B | A)$ into $P(A | B)$.

THE QUESTION

Once we learn that B happened, how does the chance of A change, and how do we turn $P(B | A)$ into $P(A | B)$?

THE INTUITION

Knowing B shrinks the sample space to B; $P(A | B)$ is the share of B where A also happens. Reversing the condition needs the prior: with 5% sick, a positive from a 90% sensitive test means about half.

KEY TAKEAWAY

$P(A | B) = P(A \cap B) / P(B)$.
Bayes' rule: posterior = likelihood × prior / evidence.
Never drop the prior.

learn B → keep only B → $P(A \cap B) / P(B)$ → multiply along a tree → add over cases → Bayes: reverse → update again

1 Conditioning on B keeps only B's outcomes

no information: all of S

1 2 3 4 5 6

given even: keep B = {2, 4, 6}

2 4 6

given six: keep {6}

6

$$P(\text{six}) = 1/6$$

1 of 6 faces

$$P(\text{six} | \text{even}) = 1/3$$

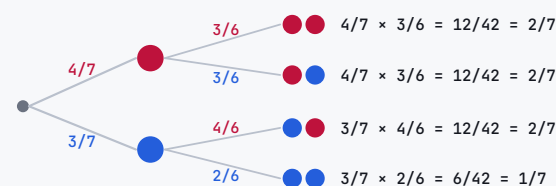
1 of the 3 kept faces

$$P(\text{even} | \text{six}) = 1$$

the only kept face is even

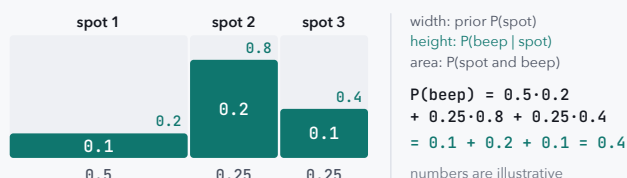
The chance of A given B is the share of B where A also happens: $P(A | B) = P(A \cap B) / P(B)$, for $P(B) > 0$; with equally likely outcomes, $|A \cap B| / |B|$. Order matters: $P(\text{six} | \text{even}) = 1/3$, but $P(\text{even} | \text{six}) = 1$. $P(\cdot | B)$ obeys the axioms, so $P(A^c | B) = 1 - P(A | B)$.

2 Multiply along the branches of a tree



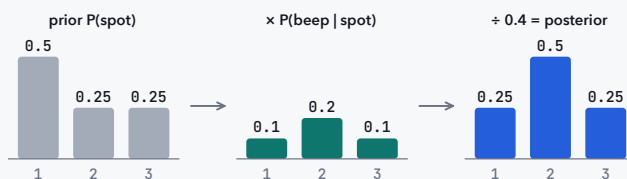
The definition rearranged is the multiplication rule, $P(A \cap B) = P(A)P(B | A)$; for three events, $P(A \cap B \cap C) = P(A)P(B | A)P(C | A \cap B)$. Each branch carries a conditional probability, a path multiplies them, and the leaves sum to 1. Two balls from 4 red and 3 blue, no replacement: $P(\text{both red}) = 2/7$, as counted on the counting sheet. The second ball is red with $2/7 + 2/7 = 4/7$, like the first.

3 Total probability adds over the cases



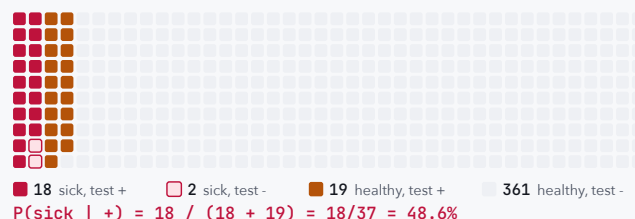
If $A_1, \dots, A_k$ split S (disjoint, together all of S), then $P(B) = \sum_i P(B | A_i)P(A_i)$, an average of the conditional chances weighted by the priors. A robot is in spot 1, 2 or 3 with prior 0.5, 0.25, 0.25; its wall sensor beeps with probability 0.2, 0.8, 0.4 there. A beep has total probability 0.4. When two groups weight the cases differently, this sum can reverse a comparison: Simpson's paradox (next page).

4 Bayes' rule turns $P(B | A)$ into $P(A | B)$



Write $P(A \cap B)$ both ways and divide: $P(A | B) = P(B | A)P(A) / P(B)$, with $P(B)$ from total probability. In words, posterior = likelihood × prior / evidence. After a beep the robot's belief moves from 0.5, 0.25, 0.25 to 0.25, 0.5, 0.25: spot 2, where beeps are likeliest, gains. Dividing by $P(\text{beep}) = 0.4$ only rescales, so the posterior sums to 1.

5 Natural frequencies: count 400 people



A test catches 90% of sick people (sensitivity) and clears 95% of healthy ones (specificity); 5% of people are sick. Of 400 people, 20 are sick and 18 of them test positive, but so do 19 of the 380 healthy. Only 18 of the 37 positives are sick, although $P(+ | \text{sick}) = 0.9$. A negative is reassuring: 2 of 363, 0.55%. Simulating 100,000 people (seed 0) gives 49.3%.

6 The prior decides what a positive means

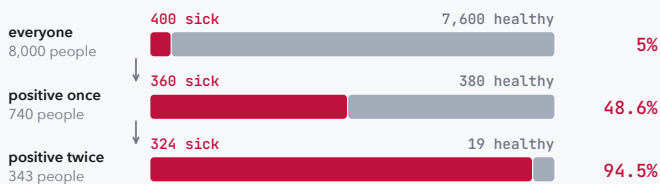


With the same test, a positive means 15.4% sick at a 1% prior, 48.6% at 5% and 81.8% at 20%. When few are sick, the false positives from the large healthy group outnumber the true ones. Specificity 0.99 lifts the 5% case to 82.6%. Reading $P(+ | \text{sick})$ as $P(\text{sick} | +)$ ignores the base rate.

Updating, independence and classic traps

Today's posterior is tomorrow's prior. Independent events leave each other's chances alone.

7 Test twice: the posterior becomes the prior



Retest the positives. Of 8,000 people, 400 are sick: 360 test positive, 324 of them again. Of the 7,600 healthy, 380 test positive, 19 of them again. So $P(\text{sick} \mid \text{two positives}) = 324/343 \approx 94.5\%$: one more Bayes step with $18/37$ as the prior. This assumes the results are independent once we know who is sick (conditional independence); a fault that repeats breaks it. A third positive gives 99.68%; a positive then a negative, 9.07%.

```
def bayes(prior, sens=0.9, spec=0.95):
    pos = (sens * prior
           + (1 - spec) * (1 - prior))
    return sens * prior / pos
bayes(0.05) # 0.4865: once
bayes(bayes(0.05)) # 0.9446: twice
```

8 Independent: knowing B leaves $P(A)$ alone

rows: 1st die, columns: 2nd die; both = dark



A and B are independent if $P(A \cap B) = P(A)P(B)$; for $P(B) > 0$ that is $P(A \mid B) = P(A)$. Check it by counting: "first die even" is independent of "sum is 7" but not of "sum is 8", where $P(A \mid C) = 3/5$. If A and B are independent, so are A and B^c . Exclusive events with positive chances are never independent (see the sets sheet). Independent, identically distributed (i.i.d.) draws multiply: six fair flips all land heads with $(1/2)^6 = 1/64$.

11 Pause and think

A fair coin gives five heads in a row. $P(\text{heads next})$? $1/2$: of the 64 six-flip sequences, 2 start with HHHHH and 1 of them ends in H. Tails is never "due".

If $P(B \mid A) > P(B)$, is $P(A \mid B) > P(A)$? Yes: both say $P(A \cap B) > P(A)P(B)$.

$P(A) = 0.5$, $P(B) = 0.4$, $P(A \cup B) = 0.7$. Independent? Yes: $P(A \cap B) = 0.2 = 0.5 \times 0.4$, and $P(A \cap B^c) = 0.3 = 0.5 \times 0.6$.

The robot's sensor stays silent. Most likely spot? Spot 1: $0.5 \times 0.8 = 0.4$ of $P(\text{silent}) = 0.6$, so $2/3$.

The host opens door 2 or 3 at random and it shows a goat. Switch? No gain: each closed door now has $1/2$. His knowledge made the $2/3$.

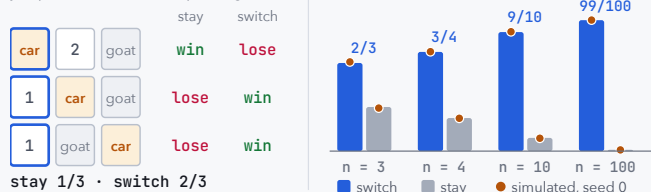
A test never flags healthy people (specificity 1). $P(\text{sick} \mid +)$? 1, whatever the prior.

$P(A \mid B) = P(A \cap B) / P(B)$ · multiply along a tree · add over cases · Bayes: likelihood $\times$ prior / evidence · independent: $P(A \cap B) = P(A)P(B)$

Source: the course's COVID testing app (Bayes' rule for a test; default prior 5%, sensitivity 0.9, specificity 0.95), the i.i.d. notebook, and course problems on Monty Hall with n doors and robot localisation, nipunbatra.github.io/psdv-teaching. The robot and teacher numbers are illustrative; the other examples and all numbers are our own, computed by the sheet's figure script; simulations use seed 0. Next: the random variables sheet.

9 Monty Hall: switching wins $2/3$

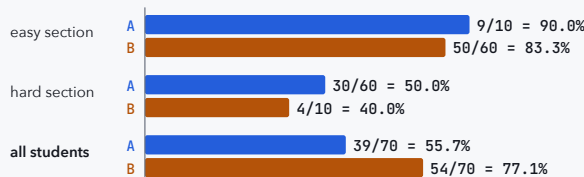
you pick door 1; the host opens a goat



You pick 1 of 3 doors; the host, who knows where the car is, opens another door with a goat. Staying wins only if your first pick was right, $1/3$. Otherwise the host must leave the car closed, so switching wins $2/3$. With n doors and $n - 2$ goats opened: stay $1/n$, switch $(n - 1)/n$, so $99/100$ at 100 doors. A simulation needs only one fact: switching wins exactly when the first pick missed.

```
rng = np.random.default_rng(0)
car, pick = rng.integers(0, 3, (2, 10**4))
(car != pick).mean() # 0.6741
```

10 Simpson's paradox: the mix flips the total



A teaches mostly the hard section (60 of 70), B mostly the easy one

Teacher A has the higher pass rate in both sections, yet B has the higher overall rate, 77.1% against 55.7%. The overall rate is total probability, $P(\text{pass} \mid A) = \sum_s P(\text{pass} \mid A, s)P(s \mid A)$ over sections s , and the weights differ: most of A's students sit the hard section. Compare within groups when a third variable (the section) affects both who is taught and who passes. Numbers are illustrative.